Question Number : 26 Question Id : 6952782161 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The density ρ of a uniform cylinder is determined by measuring its mass m, length l and diameter d. The measured values of m, l and d are 97.42 ± 0.02 g, 8.35 ± 0.05 mm and 20.20 ± 0.02 mm, respectively. Calculated percentage fractional error in ρ is _____.

Options :

6952787581. 0.63%

6952787582. 0.82%

6952787583. 0.72%

6952787584. 0.25%

Question Number : 27 Question Id : 6952782162 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The potential energy of a particle changes with distance x from a fixed origin as $V = \frac{A\sqrt{x}}{x+B}$, where A and B are constant with appropriate dimensions. The dimensions of AB are_____.

Options :

6952787585. $[M^1 L^{5/2} T^{-2}]$

6952787586. $[M^{3/2} L^{5/2} T^{-2}]$

6952787587. $[M^1 L^2 T^{-2}]$

6952787588. $[M^1 L^{7/2} T^{-2}]$

Question Number : 28 Question Id : 6952782163 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The rain drop of mass 1 g, starts with zero velocity from a height of 1 km. It hits the ground with a speed of 5 m/s. The work done by the unknown resistive force is _____J.
(take g = 10 m/s^2)

Options :

6952787589. -8.75

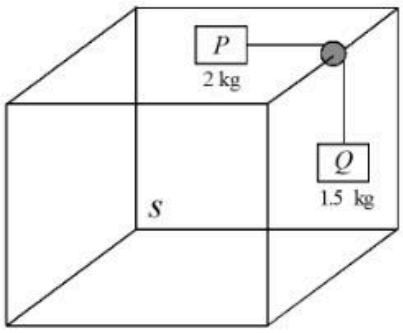
6952787590. -8.35

6952787591. -9.55

6952787592. -9.98

Question Number : 29 Question Id : 6952782164 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pully which is fixed on the edge of a cube (S), as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ. The cube (S) moves towards the right with acceleration of $\frac{g}{2}$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is _____.
(take g = 10 m/s^2)



Options :

6952787593. 0.33

6952787594. 0.67

6952787595. 1

6952787596. 0.5

Question Number : 30 Question Id : 6952782165 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is 4×10^8 N/m^2 and its radius is 4 mm, then maximum acceleration the lift can take is _____ m/s^2.
(take g = 10 m/s^2 and $\pi = 3.14$)

Options :

6952787597. 2.56

6952787598. 3.89

6952787599. 4.32

6952787600. 5.16

Question Number : 31 Question Id : 6952782166 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A solid sphere of radius 4 cm and mass 5 kg is rotating (rotation axis is passing through the centre of the sphere) with an angular velocity of 1200 rpm. It is brought to rest in 10 s by applying a constant torque. The torque applied and the number of rotations it made before it comes to rest are _____ and _____ respectively.

Options :

6952787601. $0.128\ \pi$ Nm, 100

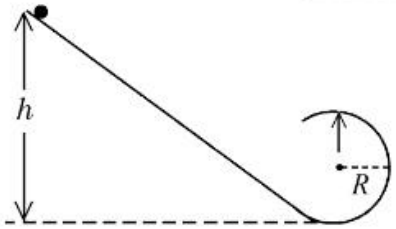
6952787602. $0.0128\ \pi$ Nm, 50

6952787603. $0.128\ \pi$ Nm, 50

6952787604. $0.0128\ \pi$ Nm, 100

Question Number : 32 Question Id : 6952782167 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height h as shown. If the body exerts a force of three times its weight on the plane at the highest point of circle then the height $h = \alpha R$. The value of α is _____.



Options :

6952787605. 2

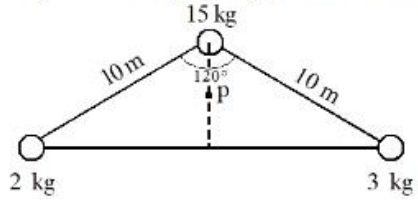
6952787606. 4

6952787607. 3

6952787608. 6

Question Number : 33 Question Id : 6952782168 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The position of center of mass of three masses 2 kg, 3 kg and 15 kg placed with respect to mid point (p) of normal bisector, as shown in the figure is_____.



Options :

6952787609. $\left(\frac{\sqrt{3}}{4}, 1.25\right)$

6952787610. $\left(\frac{\sqrt{3}}{4}, 1.0\right)$

6952787611. $(0,0)$

6952787612. $(1.25,0)$

Question Number : 34 Question Id : 6952782169 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The two wires A and B of equal cross-section but of different materials are joined together. The ratio of Young's modulus of wire A and wire B is 20/11. When the joined wire is kept under certain tension the elongations in the wires A and B are equal. If the length of wire A is 2.2 m, then the length of wire B is _____m.

Options :

6952787613. 1.1

6952787614. 2.22

6952787615. 1.21

6952787616. 4.44

Question Number : 35 Question Id : 6952782170 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K. Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K. The final pressure in the vessels is _____kPa.

Options :

6952787617. 100

6952787618. 120

6952787619. 90

6952787620. 105

Question Number : 36 Question Id : 6952782171 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In interference experiment the path difference between two interfering waves at a point A on the screen is $\lambda/3$, where λ is the wavelength of these waves, and at another point B the path difference is $\lambda/6$. The ratio of intensities at points A and B is_____.

Options :

6952787621. 3

6952787622. 4

6952787623. 1/3

6952787624. 1/4

Question Number : 37 Question Id : 6952782172 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A particle is executing simple harmonic motion. Its amplitude is A and time period is 5 sec. The time required by it to move from $x = A$ to $x = \frac{A}{\sqrt{2}}$ is _____ sec.

Options :

6952787625. 1/4

6952787626. 5/4

6952787627. 5/8

6952787628. 3/8

Question Number : 38 Question Id : 6952782173 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A thin half ring of radius 35 cm is uniformly charged with a total charge of Q coulomb. If the magnitude of the electric field at centre of the half ring is 100 V/m, then the value of Q is _____ nC.

($\epsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2 and $\pi = 3.14$)

Options :

6952787629. 2.14

6952787630. 2.44

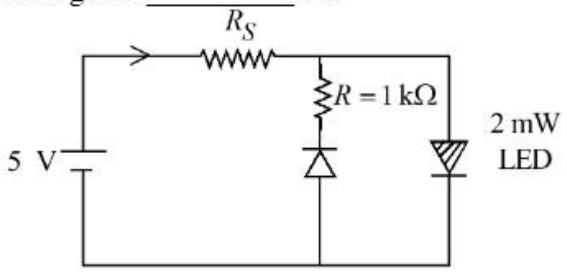
6952787631. 3.25

6952787632. 0.7

Question Number : 39 Question Id : 6952782174 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The maximum rated power of the LED is 2 mW and it is used in the circuit with input voltage of 5 V as shown in the figure below. The current through resistance R_S is 0.5 mA.

The minimum value of the resistance of R_S, to ensure that the LED is not damaged is _____ kΩ.



Options :

6952787633. 6

6952787634. 2

6952787635. 4

6952787636. 5

Question Number : 40 Question Id : 6952782175 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A point light source emits E.M. waves in free space. A detector, placed at a distance of L m, measures the intensity as I_o. The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45°. The measured intensity at new location will be _____.

Options :

6952787637. $\frac{I_o}{4}$

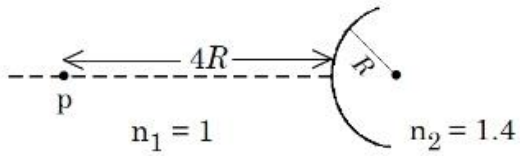
6952787638. I_o

6952787639. $\frac{I_o}{\sqrt{2}}$

6952787640. $\frac{I_o}{2}$

Question Number : 41 Question Id : 6952782176 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of $4R$ in front of spherical interface. The magnitude of the magnification of point source image is _____.



Options :

6952787641. 1.66

6952787642. 2.33

6952787643. 2.66

6952787644. 1.33

Question Number : 42 Question Id : 6952782177 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A small cube of side 1 mm is placed at the centre of a circular loop of radius 10 cm carrying a current of 2 A. The magnetic energy stored inside the cube is $\alpha \times 10^{-14}$ J. The value of α is _____.
($\mu_o = 4\pi \times 10^{-7}$ Tm/A, $\pi = 3.14$)

Options :

6952787645. 6.28

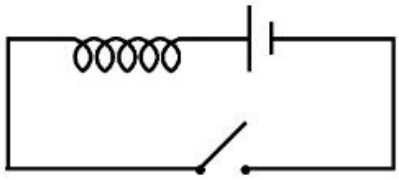
6952787646. 6.28×10^{-6}

6952787647. 628

6952787648. 6.28×10^{-4}

Question Number : 43 Question Id : 6952782178 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An inductor of inductance 10 mH having resistance of 100 Ω is connected to battery of E.M.F. 1.0 V through a switch as shown in the figure below. After switch is closed, the ratio of instantaneous voltages across the inductor when the current passing through it is 2 mA and 4 mA is _____.



Options :

6952787649. 4/3

6952787650. 3/4

6952787651. 5/3

6952787652. 3/5

Question Number : 44 Question Id : 6952782179 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β. The possible values of α and β are:-

Options :

6952787653. 27 and 20

6952787654. 3 and 16

6952787655. 5 and 36

6952787656. 20 and 27

Question Number : 45 Question Id : 6952782180 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A LCR series circuit driven with E_{rms} = 90 V at frequency f_d = 30 Hz has resistance R = 80 Ω, an inductance with inductive reactance X_L = 20.0 Ω and capacitance with capacitive reactance X_C = 80.0 Ω. The power factor of the circuit is_____.

Options :

6952787657. 0.8

6952787658. 0.64

6952787659. 0.9

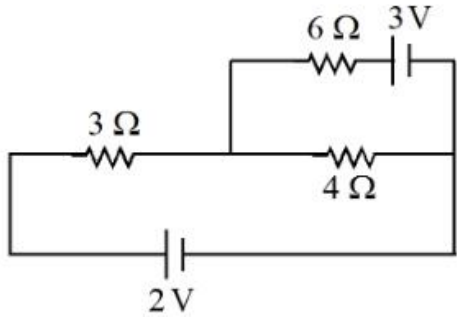
6952787660. 0.5

Physics Section B

Section Id :	695278154
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	695278154
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Id : 6952782181 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Refer to the circuit diagram given below. The heat generated across the 6 Ω resistance in 100 second is $\frac{\alpha}{100}$ J . The value of α is _____. (Nearest integer)



Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 47 Question Id : 6952782182 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

An unpolarized light of intensity I_0 passes through polarizer and then through a certain optically active solution and finally it goes to analyser. If the angle between analyser and polariser is 0° and intensity of light emerged from analyser is $\frac{3}{8}I_{0,}$ the angle of rotation of the light by the solution with respect to analyser is _____ degrees.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 48 Question Id : 6952782183 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The energy released when $\frac{7}{17.13}$ kg of $^{7}_{3}Li$ is converted into $^{4}_{2}He$ by proton bombardment is $\alpha \times 10^{32}$ eV. The value of α is ____. (Nearest integer)
(Mass of $^{7}_{3}Li$ = 7.0183 u, mass of $^{4}_{2}He$ = 4.004 u, mass of proton = 1.008 u and 1 u = 931 MeV/c^2 and Avogadro number = 6.0×10^{23})

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 49 Question Id : 6952782184 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A three coulomb charge moves from the point (0, −2, −5) to the point (5, 1, 2) in an electric field expressed as $\vec{E} = 2x\hat{i} + 3y^2\hat{j} + 4\hat{k}$ N/C. The work done in moving the charge is _____J.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 50 Question Id : 6952782185 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A certain gas is isothermally compressed to $\left(\frac{1}{3}\right)^{rd}$ of its initial volume (V_o = 3 litre) by applying required pressure. If the bulk modulus of the gas is 3×10^5 N/m^2, the magnitude of work done on the gas is _____J.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	695278155
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	695278155
Question Shuffling Allowed :	Yes
Is Section Default? :	No